

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

B. Tech IG (Civil Engineering) SEMESTER - 3 Winter 2025 (Regular)

Course :Bachelor of Technology (Integrated) (Civil Engineering (Structural Engineering)) Branch : Engineering and Technology

Semester : SEMESTER - 3

Subject Code & Name: 25AF1906PC302 - MECHANICS OF SOLIDS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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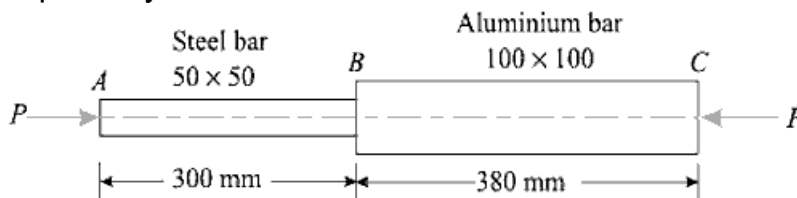
Q1. Objective type questions. (Compulsory Question) 12

- 1 A mild steel wire 5 mm in diameter and 1 m long. If the wire is subjected to an axial tensile load of 10 kN, what will be its extension?
a) 3.15 mm b) 2.55 mm c) 2.65 mm d) 2.45 mm
- 2 A 1.5 mm thick sheet is subject to unequal biaxial stretching, and the true strains in the directions of stretching are 0.05 and 0.09. The final thickness of the sheet in mm is
(A) 1.414 (B) 1.304 (C) 1.362 (D) 289
- 3 The yield stress of a metal in uniaxial tension is 200 MPa. According to von Mises yield criterion, the yield stress (in MPa) of the metal in pure shear is closest to.....
a) 115.5 b) 100 c) 66.70 d) 141.40
- 4 Which of the following materials is most likely to fail in shear?
a) Wood b) Concrete c) Rubber d) Steel
- 5 When SF at a point is zero, then BM at that point will be:
(A) Zero (b) Minimum (c) Maximum (d) Infinity.
- 6 The point of contraflexure is a point where:
(A) SF changes sign (B) BM changes sign
(C) SF is maximum (D) BM is maximum.
- 7 Neutral axis of a section is an axis at which the bending stress is:
(A) Minimum (B) zero (C) Maximum (D) Infinity
- 8 In the theory of simple bending, the bending stress in the beam section varies:
(A) linearly (B) parabolically (C) elliptically (D) none
- 9 When a cantilever is loaded at its free end, maximum compressive stress shall develop at:
(A) Bottom fibre (B) Top fibre (C) Neutral axis (D) Centre of gravity
- 10 The effective length of a column hinged at both ends
(A) L (B) 0.7 L (C) 1.414 L (D) 2 L

- 11 The slenderness ratio of a long column is:
 (A) 10 – 20 (B) 20 – 30 (C) 50 – 60 (D) above 80
- 12 When a rectangular section of a beam is subjected to a shearing force, the ratio of maximum shear stress to average shear stress is:
 (A) 2.0 (B) 1.75 (C) 1.5 (D) 1.25

Q2. Solve the following.

- A) An alloy wire of 2 mm^2 cross-sectional area and 12 N weight hangs freely under its own weight. Find the maximum length of the wire if its extension is not to exceed 0.6 mm. Take E for the wire material as 150 GPa.
- B) A member formed by connecting a **steel bar** to an **aluminum bar** is shown in **the fig.** below. Assuming that the bars are prevented from buckling sideways, calculate the magnitude of force P that will cause the total length of the member to decrease by 0.25 mm. The values of elastic modulus for steel and aluminum are 210 GPa and 70 GPa, respectively.



Q3. Solve the following.

- A) A simply supported beam 6 m long is carrying a uniformly distributed load (UDL) of 5 kN/m over a length of 3 m from the right end. Draw the S.F. and B.M. diagrams for the beam, and also calculate the maximum B.M. on the section.
- B) Draw the SF and BM diagrams for a simply supported beam of length L carrying a UDL of ' w ' kN/m throughout the length.

Q4. Solve Any Two of the following.

- A) A cantilever beam is rectangular in section, having 80 mm width and 120 mm depth. If the cantilever is subjected to a point load of 6 kN at the free end and the bending stress is not to exceed 40 MPa, find the span of the cantilever beam.
- B) A wooden beam 100 mm wide, 250 mm deep, and 3 m long is carrying a uniformly distributed load of 40 kN/m. Determine the maximum shear stress and sketch the variation of shear stress along the depth of the beam.
- C) A solid steel shaft is to transmit a torque of 10 kN-m. If the shearing stress is not to exceed 45 MPa, find maximum diameter of the shaft.

Q5. Solve Any Two of the following.

- A) What are the assumptions & limitations of Euler's column theory?
- B) A hollow alloy tube of 4 m length with an external diameter of 40 mm and an internal diameter of 25 mm was found to extend 4.8 mm under a tensile load of 60 kN. Find the buckling load for the tube with both ends pinned. Also, find the safe load on the tube, taking a factor of safety of 5.

- C) Explain in detail the term "**equivalent length of a column**" and the relation between equivalent length (L_e) and actual length (L). 6

Q6. Solve Any Two of the following.

- A) Define the principal planes & principal stresses & explain their uses. 6
- B) The stresses at a point of a machine component are 150 MPa and 50 MPa, both tensile. Find the intensities of normal stress, shear stress, and resultant stress on a plane inclined at an angle of 55° with the axis of major tensile stress. Also, find the magnitude of the maximum shear stress in the component.
- C) Write a short note on **the maximum principal stress theory**. 6

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